

DESIGN

Helix OTA — Emulation/Virtualization Test-Fabric — Architecture & Design

Field	Value
Revision	2
Last modified	2026-06-11T10:30:00Z
Status	active — T0 floor + Tab dev-host A/B-virt FOUNDATION built; Tab slot mechanism in progress The incorporation design for a LOCAL + DISTRIBUTED hardware-free test fabric, derived from the cited verdicts in ../.. /research/emulation_infra/REPORT.md and the dev-host A/B-virt work in ../.. /research/rk3588_emulator/REPORT.md. Extends the containers submodule (§11.4.76) and the existing Tier-1/2/3 baseline in ../EMULATED_DEVICE_TESTING.md.
Status summary	Reconciles the macOS-M3 dev host vs the Linux-KVM CI tier as FACT (§11.4.6/§11.4.112). Now partially built: the T0 floor (shipped) plus a new Tab dev-host A/B-virt tier (QEMU virt+HVF + U-Boot bootcount/altbootcmd + RAUC dm-verity) whose FOUNDATION boots to live userspace (PROVEN); its real A/B slot switch / dm-verity / auto-rollback is in progress, gated on the in-flight u-boot.bin build — NOT proven (§11.4.6).
Authority	Operator mandate 2026-06-10 (hardware-free comprehensive emulation/VM/container fabric)
Related	../.. /research/emulation_infra/REPORT.md; ../.. /research/rk3588_emulator/REPORT.md;

Field	Value
	TEST_COVERAGE_PLAN.md; SCHEMA.sql; ROADMAP.md; containers/ submodule

1. Goal & non-goals

Goal. A single fabric that boots/runs emulated, virtual, and (where wired) real targets — locally on the dev host AND distributed across remote/CI nodes — so every Helix OTA work item is testable WITHOUT a physical RK3588/Orange Pi on the operator’s desk. Anti-bluff is the spine: a tier never claims fidelity it does not have (§11.4 / §11.4.107), and the tier boundaries below are stated as FACT precisely so no tier silently over-claims.

Non-goals. (a) Reinventing container/VM machinery — every primitive is an EXTENSION of vasic-digital/containers (§11.4.74/§11.4.76), never a re-implementation. (b) Pretending the macOS-M3 host can run KVM-gated Android — it cannot (§3); that is an honest host gap (§11.4.112), not a thing to fake.

2. Tier model (reconciled with EMULATED_DEVICE_TESTING.md)

Tier	Fidelity	Tech (from REPORT verdicts)	Where it runs	What it proves
T0 — protocol client	wire/server flow only (no real apply)	existing ota-device-emulator (server/internal/deviceemu) on podman	dev host + anywhere	register→update-check→telemetry→delta→rollout→ against a live control plane
T1 — Android userspace	real Android app/agent logic; NOT real A/B apply	Android AVD arm64-v8a (HVF local / KVM CI); Genymotion optional	dev host (HVF, accelerated) + Linux/ KVM CI	ota-android-agent/bridge decision logic, ADB-driven UI/automation
Tab — dev-host A/B-virt (built; slot mechanism in progress)	real U-Boot bootcount/ altbootcmd A/B slot-switch + RAUC dm-verity + auto-rollback on a generic aarch64 virt board (NOT the RK3588	QEMU -mac hine virt + HVF aarch64 guest (Buildroot) + U-Boot + RAUC (tests/emulator/ab_virt/)	dev host (HVF, accelerated)	the real A/B slot-switch + rollback semantics on this Apple-Silicon host WITHOUT a KVM gate — closes p the T0→T2 gap locally. FOUNDAT boots to live userspace (PROVEN) slot switch itself is authored-not-ye run (§11.4.6, gated on the u-boot. build).

Tier	Fidelity	Tech (from REPORT verdicts)	Where it runs	What it proves
T2 — high-fidelity A/B	SoC, NOT Android update_engine) real update_engine A/B + AVB/dm-verity + auto-rollback on virtual partitions	Cuttlefish (c vd)	Linux+KVM CI only (host-gated on M3)	the genuine Android OTA apply + rollback semantics short of real silicon
T3 — real hardware	vendor HAL + real U-Boot slot-switch + dm-verity on real partitions + Mali GPU	RK3588/ Orange Pi 5 Max behind LAVA; redroid-rk3588 adjacent (SoC app/GPU, no bootloader)	HIL lab	the only env exercising real silicon; flashes governed by §11.4.133
Tfw — firmware/Linux-target	aarch64 UEFI/U-Boot/Linux boot (generic virt board, not RK3588 SoC)	QEMU -machine virt + edk2 (existing pkg/vm); Renode future	dev host (TCG) + CI	U-Boot slot logic, future Linux/RTOS targets
Tcp — control-plane isolation	N/A (not a device)	Firecracker / Kata / Cloud Hypervisor microVMs; gVisor sandbox	Linux+KVM CI (gVisor needs no KVM)	scaling/DDoS/chaos/security of the control plane under real isolation

Cross-cutting: **distributed fan-out** via **LAVA** (one job schema drives BOTH virtual T0/T1/T2 and physical T3) and/or **Nomad** (heterogeneous scheduler for a mixed container+QEMU+cvd fabric); **CI front door** via **our own** self-hosted mac+Linux runners (GitHub-hosted Apple-Silicon runners do NOT expose HVF/nested-virt — FACT, REPORT §5).

flowchart TB

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    subgraph DEV["Dev host – Apple M3 / macOS (HVF, no /dev/kvm)"]
        T0d["T0 ota-device-emulator (podman)"]
        T1d["T1 Android AVD arm64-v8a (HVF)"]
        TABd["Tab A/B-virt: QEMU virt+HVF + U-Boot + RAUC (foundation booted; slot switch in progress)"]
        Fwd["Tfw QEMU virt + edk2 / Renode"]
    end
    subgraph CI["Distributed CI – Linux + KVM nodes"]
        T2["T2 Cuttlefish (cvd) – real A/B/AVB/dm-verity"]
    end

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T0c["T0 emulator fan-out (podman)"]
T1c["T1 AVD/KVM farm"]
TCP["Tcp Firecracker/Kata microVMs – CP scaling/chaos"]
end
subgraph HIL["HIL lab – real silicon"]
  T3["T3 RK3588 / Orange Pi 5 Max"]
  RD["redroid-rk3588 (SoC app/GPU)"]
end
CP["Helix OTA control plane (server/)"]
SCHED["Fabric scheduler – LAVA job schema + Nomad/K8s exec"]
RUN["CI front door – self-hosted mac+Linux runners"]
RUN --> SCHED
SCHED --> DEV & CI & HIL
T0d & T1d & TABd & T2 & T0c & T1c & T3 & RD -->|ota-protocol| CP
SCHED -->|exclusive lease per target §11.4.119| T2 & T3

```

3. The host fact (FACT, §11.4.6 — drives every placement)

Dev host = **Apple M3 Pro / macOS 15.5 / arm64**, accel = **HVF**, **no /dev/kvm**, and **M3 lacks hardware nested-virt** (M4+/macOS-15 only) — so KVM-gated Android (Cuttlefish, redroid, K8s-STF emulator nodes, Firecracker/Kata/Cloud-Hypervisor) is **host-gated here, NOT structurally impossible** (runs on a Linux-KVM box / M4+ / GCE nested-virt). The fabric therefore places T2/Tcp on the **Linux+KVM CI tier**, and keeps T0 + T1(AVD-HVF) + Tfw(QEMU-TCG) as the **locally-runnable** set. This split is asserted, never wished away.

4. Extension of the containers submodule (§11.4.74 extend-don't-reimplement)

Reuse (already present): pkg/boot + pkg/compose + pkg/health (on-demand boot + readiness — the §11.4.76 on-demand-infra invariant the pgx integration test already uses), pkg/emulator (AVD x86_64/arm64 + HVF/KVM accel gating + AVD-lock/orphan-qemu reaping), pkg/vm (QEMU aarch64 virt+UEFI), pkg/genymotion.

Tab dev-host A/B-virt (built — tests/emulator/ab_virt/): the Tab tier reuses **U-Boot** (bootcount/altbootcmd slot-select) and **RAUC** (dm-verity A/B update client) upstream per §11.4.74 — neither reimplemented in-project — on top of pkg/vm's QEMU aarch64 virt+HVF boot. The guest image (Buildroot base + kernel) is built inside a named podman aarch64 Linux container (this host is macOS); the 2-slot GPT disk assembler (assemble_ab_disk.sh) + U-Boot boot.cmd/env (uboot_ab/) are authored. The base image build (PWU-AB-0) and the live-userspace boot (PWU-AB-1 FOUNDATION) are DONE+PROVEN; the slot-switch disk is gated on the in-flight u-boot.bin build (§11.4.6 — in progress).

New primitives to ADD upstream (PR to vasic-digital/containers), never copied in-project: - pkg/cuttlefish — cvd lifecycle wrapper (fetch/launch/stop, aosp_cf_arm64_only_phone, KVM-presence gate that SKIPS-with-reason on a non-KVM host). - pkg/lava — LAVA REST/XML-RPC client: submit a job, poll,

pull artifacts; one job schema for virtual + physical DUTs. - pkg/fabric — the distributed scheduler façade + **target registry** (lease/release a target exclusively, §11.4.119) over a Nomad/K8s/LAVA backend.

Stays in helix_ota (OTA-domain logic, not generic plumbing): server/internal/deviceemu (the T0 emulator — it speaks ota-protocol), the per-tier on-device test drivers, and the bank/Challenge wiring.

5. On-demand boot + single-resource-owner contracts

- **On-demand boot (§11.4.76):** the test entry point boots its tier's infra via the submodule (pkg/boot/compose/health/fabric); operators never hand-start podman, an emulator, cvd, or a LAVA job. A tier with unmet host prereqs **SKIPS-with-reason** (§11.4.3), never fakes a PASS.
- **Single-resource-owner (§11.4.119):** every exclusive target (a cvd instance, a real board, an HDMI/serial line, a fixed port) is leased to **exactly one** driver stream at a time via pkg/fabric's registry; other streams targeting it are read-only or queued. Concurrent drivers of one exclusive target produce cross-contaminated evidence = a §11.4 bluff, so the lease is mandatory.

6. Persistence

A test-fabric registry (targets, runs, results, evidence paths, distributed-node inventory) is defined in `SCHEMA.sql`. It is OPTIONAL for local single-node use (the T0 in-process/podman tiers need no DB) and REQUIRED once distributed fan-out is wired (so the scheduler can lease targets and the dashboard can show fleet-of-fabric state). It reuses the project's pgx/Postgres seam, not a new datastore.

7. Anti-bluff posture (the spine)

Each tier produces captured evidence under docs/qa/<run-id>/ per `TEST_COVERAGE_PLAN.md`: T0 captures real request/response transcripts; T1 captures ADB/UI + update_engine_client state; **Tab** captures the QEMU+HVF boot console (docs/qa/20260611T061626Z-ab-virt-boot/console.log, HELIX_USERSPACE_LIVE_OK — the FOUNDATION proof) and, once its disk is runnable, the slot-state assertion (findmnt / + cat /etc/slot_id across a switch) + the auto-rollback trace; T2 captures real update_engine A/B-slot + AVB/dm-verity + auto-rollback evidence; T3 adds on-silicon apply/rollback. A tier claiming a PASS without exercising the real system at its fidelity is a §11.4 PASS-bluff. Honest gaps (§3, REPORT §8) are documented, never hidden behind a green line — the Tab slot switch is recorded as **NOT proven** (gated on u-boot.bin) and T2 stays **host-gated** (no KVM on this macOS host), never as faked greens.